Lab 1 Report

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1. Problem and Assumptions

In this lab I did three things. First I made a timeline of AI history. Second I wrote a rule based scholarship checker in Python, added two new rules to it and tested it with four cases. Third I wrote why passing a conversational test does not prove general intelligence. No machine learning model is trained in this lab.

All numbers used in the lab (score >= 70, attendance >= 75, composite >= 75, weights 0.6 and 0.4) are synthetic lab assumptions. They are not a real scholarship policy. All the students in the data (A01 to A12) are made up.

I used two Python files. lab1_rules.py has the rule checker and the four tests. lab1_compare.py compares the rule checker with a weighted score method on 12 made up students from applicants.csv. Both files only use the standard Python library.

2. Timeline of AI Milestones

1950 - Turing and the imitation game

Alan Turing wrote "Computing Machinery and Intelligence" and proposed the imitation game. It changed the question "can machines think" into a test based on behaviour. Source. Turing (1950).

1956 - Dartmouth Summer Research Project

The project was proposed in 1955 and held in 1956. The term artificial intelligence was introduced here and it is seen as the start of AI as a field. Source. McCarthy et al. (1955).

c. 1974 to 1980 - First AI winter

Funding and interest in AI dropped after the Lighthill report in 1973. It shows that AI did not progress in a straight line, big promises were followed by funding cuts. Source. Lighthill (1973); Crevier (1993).

2012 - AlexNet (modern compute and data)

AlexNet won the ImageNet competition using GPUs and the large ImageNet dataset. It showed that big data and GPU power can make deep neural networks much better than older methods. Source. Krizhevsky et al. (2012); Deng et al. (2009).

2017 - The Transformer

The Transformer model was introduced in the paper "Attention Is All You Need". It is the base of most modern language models and chatbots. Source. Vaswani et al. (2017).

3. Rule Based Checker

The original rules were score >= 70 and documents complete. I added two named rules. Minimum.attendance (attendance must be at least 75%) and clean.conduct (the student

must have no disciplinary record). A student is ELIGIBLE only if all four rules pass. Again, the thresholds are synthetic lab assumptions.

I chose exactly four test cases. T1 is a normal pass. T2 breaks only the attendance rule and T3 breaks only the conduct rule, with everything else correct, so I know which rule caused the failure. T4 is a boundary case where the score is exactly 70, to check that the rule uses >= and not >.

T1 normal pass

- Input. Score 85, documents complete, attendance 90, no record
- Expected. ELIGIBLE. Actual. ELIGIBLE. Reason. All rules passed

T2 attendance rule (minimum.attendance)

- Input. Score 85, documents complete, attendance 60, no record
- Expected. NOT ELIGIBLE. Actual. NOT ELIGIBLE. Reason. Attendance 60% is less than 75%

T3 conduct rule (clean.conduct)

- Input. Score 85, documents complete, attendance 90, has disciplinary record
- Expected. NOT ELIGIBLE. Actual. NOT ELIGIBLE. Reason. Student has a disciplinary record

T4 boundary

- Input. Score 70, documents complete, attendance 75, no record
- Expected. ELIGIBLE. Actual. ELIGIBLE. Reason. All rules passed

All four tests gave the expected result.

4. Comparison. Rules vs Weighted Score

I compared two methods. Method A is the hard rules, where every rule must pass. Method B is a weighted score, where composite = 0.6 x score + 0.4 x attendance and it must be at least 75. In method B documents and conduct are still must-have conditions. Out of 12 students the two methods gave the same answer for 8 and a different answer for 4.

- A02 (score 70, attendance 75). Hard rules say ELIGIBLE, weighted score says NOT ELIGIBLE (composite 72.0).
- A03 (score 95, attendance 70). Hard rules say NOT ELIGIBLE because attendance is below 75, weighted score says ELIGIBLE (composite 85.0).
- A04 (score 65, attendance 100). Hard rules say NOT ELIGIBLE because score is below 70, weighted score says ELIGIBLE (composite 79.0).
- A09 (score 90, attendance 74). Hard rules say NOT ELIGIBLE because attendance is below 75, weighted score says ELIGIBLE (composite 83.6).

The hard rules are easy to explain because the reason for rejection is clear, but they are strict at the cut-off. The weighted score is more flexible, but the weights are chosen by me and a high score in one area can hide a weak point in another, like A04. Limitations. I only used 12 made up students and there is no correct answer to check against, so I cannot say

which method is better. Attendance and conduct can also be unfair to some students, for example if they were ill, so a real system should have a way to review or appeal.

5. Why Passing a Conversational Test Is Not General Intelligence

Take the scholarship case where a student chats with an assistant to ask if they qualify. The conversational test (the imitation game from Turing, 1950) only looks at the text conversation. It checks if a person can tell the machine from a human. So in this case it only observes that the assistant answers in a fluent and believable way.

It does not prove that the answer is correct. It also does not prove that the machine understands anything. Searle (1980) argued with the Chinese Room example that giving the right looking output is not the same as understanding. ELIZA (Weizenbaum, 1966) is an older example where a very simple program made people feel it understood them. Also, doing well in one chat does not show that it will work on other tasks.

Example of fluent but wrong reasoning. A student says "my score is 69, my documents are complete, my attendance is 90 and I have no record, do I qualify?" The assistant replies "Yes, you qualify, 69 rounds to 70 and your good attendance covers the difference." This sounds confident, but the rule says score >= 70, so the right answer is NOT ELIGIBLE. The assistant made up a rounding rule that does not exist.

One more test for reliable task performance. Make about 50 new student records, including boundary values like score 69, 70, 71 and attendance 74, 75, 76, and work out the correct answers from the written rules. Ask the system to decide each one several times with differently worded questions. Then check how many answers are correct, whether the same case always gets the same answer, and whether the reason given is the rule that really failed.

6. Failure / Edge Case

Input. Student A09 with score 90, documents complete, attendance 74% and no disciplinary record.

Observed behaviour. The hard rule checker says NOT ELIGIBLE because 74 is less than 75. The weighted method says ELIGIBLE with a composite of 83.6.

Likely cause. The attendance rule is a strict cut-off, so being 1 point short gives the same result as being far below. The student also has a very high score, which the rules do not take into account. The 75% limit is also just an assumption I chose.

Proposed improvement. Add a borderline range (for example attendance 72 to 74) that gives REVIEW instead of a final rejection, so a person can check the case. The thresholds should also be matched to a real policy.

7. Reflection and Responsible Practice

[Write your reflection here in your own words. For example, what you found easy or difficult, and what you learned from the disagreements between the two methods.]

- All data is made up. No real student information or passwords are uploaded to GitHub.

- The thresholds are marked as lab assumptions so they are not mistaken for real rules.
- For real decisions about people, a person should check the result and the student should be able to appeal.
- Historical facts are supported by the sources below. No outside code was copied.

8. How to Run

Python 3 is needed, no extra packages. Go to the lab1 folder and run. python lab1_rules.py and then python lab1_compare.py. The input file is applicants.csv and the saved outputs are in the outputs folder.